

2. (a) Magnesium reacts with oxygen to form magnesium oxide.

Using the electronic structures below, draw dot and cross diagrams to show how bonding takes place during the formation of magnesium oxide. Include the electronic structures of the ions formed. [3]

magnesium = 2,8,2

oxygen = 2,6

- (b) The melting points of magnesium oxide and sodium chloride are given below.

Substance	Melting point (°C)
sodium chloride	801
magnesium oxide	2852

Explain why there is a difference in their melting points even though they are both ionic substances. [2]

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- (c) Using the electronic structures below, draw a dot and cross diagram to show the bonding in a molecule of carbon dioxide, CO_2 . [2]

carbon = 2,4

oxygen = 2,6

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